

Syed Fuzail Farhan

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Education

University of Waterloo — BAsC, Mechatronics Engineering (Co-op)

Technical Skills

Embedded and Control: STM32 (HAL, CubeMX), ESP32 and ESP32-C6, Arduino, Raspberry Pi, FreeRTOS, micro-ROS, PWM, ADC

Protocols: UART, SPI, I2C, CAN, Ethernet, TCP/IP, WebSocket, ESP-NOW

Motor and Power: 3-phase BLDC field-oriented control, DRV8353 and DRV8833 drivers, stepper and servo drive, FPGA (Quartus, VHDL)

Mechanical: SolidWorks, AutoCAD, GD&T, ANSYS Mechanical and CFD, AGMA analysis, 3D printing, CNC machining

Software: C, C++, Python, MATLAB, Django, OpenCV, YOLO, Git, Docker, Linux

Experience

Research Assistant, IDEAS Clinic — University of Waterloo

May 2026 – Aug 2026

- Integrated a robotic pick-and-place cell from concept to commissioning using a Dobot arm and Orbbec depth camera, writing an OpenCV colour-segmentation pipeline that sorts parts without operator intervention.
- Eliminated the pinch hazard on a robotic work cell by adding a YOLO hand detector that halts arm motion within one frame of a hand entering the work envelope and holds it stopped until the zone clears.
- Enabled collision-free coordination of a TETRIX robot fleet by writing a central arbiter that exchanges newline-delimited JSON with each robot over TCP, with a dashboard streaming per-robot pose, state and sensor telemetry.
- Root-caused recurring drivetrain failures across a fleet of robot kits by running AGMA hand calculations against ANSYS Discovery and Mechanical contact FEA, measuring 139 MPa root bending and 883 MPa contact stress on a 30-tooth spur pair, and specified replacement components sized to the required duty cycle.

Firmware Developer, UW Robotics Team (Mars Rover)

June 2026 – Present

- Brought micro-ROS (Humble) up on an STM32 NUCLEO-F439ZI in STM32CubeIDE, building the 69-package 7.4 MB static library and clearing LwIP and Docker toolchain failures that were blocking the firmware team, reaching a two-node publish/subscribe demo.
- Migrating rover arm-to-base communications to Ethernet across two NVIDIA Jetsons and an STM32 controller to remove the bandwidth ceiling on the previous link.

Research Assistant, IDEAS Clinic — University of Waterloo

Aug 2025 – Dec 2025

- Co-designed a 3-day ANSYS FEA/CFD workshop delivered to 70+ students, built around industry case studies developed with ANSYS covering heat exchangers, schlieren flow imaging, a shoulder implant and a hands-free crutch.
- Validated simulation results physically by designing test models in SolidWorks, fabricating them by 3D printing, and correlating measured results against simulation to within roughly 80%.
- Designed a 4-day microcontroller and FPGA workshop for 60+ students covering PWM, serial communication and digital I/O in Arduino and Quartus Prime, including circuit debugging with multimeters and oscilloscopes.

Firmware Engineer, Electrium Design Team

Sep 2024 – Dec 2024

- Built an electronic speed controller for a personal electric vehicle on ESP32 using ESP-IDF in PlatformIO, implementing 3-phase BLDC field-oriented control.
- Configured DRV8353 gate drivers to switch a high-current MOSFET stage, tuning dead time and drive strength on the bench with an oscilloscope.

Software Engineering Intern, GrazeAI

May 2024 – Aug 2024

- Cut clinical trial identification from multiple days to minutes by engineering a reusable Python scraper over REST APIs and regex that filters 600,000+ trials against custom criteria.
- Led a team of 4 building generalised data-extraction systems, and reduced pipeline execution time by 40% through profiling and refactoring.